

Factor Graph–Based Tightly Coupled PPP-B2b/INS for Real-Time Precise Positioning

Ruite Yi , Xiangwei Zhu , Mingjun Ouyang, Chengchao Bai , and Guangteng Fan 

Abstract—Real-time precise point positioning (PPP) based on BDS-3 PPP-B2b corrections often suffers from limited robustness in complex environments. To address these challenges, this letter proposes a factor-graph-based tightly coupled PPP-B2b/inertial navigation system (INS) integration scheme. Unlike traditional filtering approaches, the proposed framework fuses asynchronous GNSS observations corrected by PPP-B2b with pre-integrated inertial measurements within a sliding-window optimization architecture. To explicitly mitigate non-Gaussian noise and measurement anomalies prevalent in urban canyons, a robust estimation strategy is implemented. This strategy incorporates Huber loss functions for outlier resistance and, critically, employs a time-varying, correction-age-dependent measurement covariance model to characterize the reliability degradation caused by correction latency. This formulation enables multiple relinearizations and effective outlier mitigation while maintaining bounded computational cost. Experimental results utilizing a dense urban dataset demonstrate that the proposed method reduces the ENU RMS position errors from 0.797/0.954/1.616 m to 0.490/0.671/0.797 m and the 3D RMS from 2.04 m to 1.15 m, representing a 43.6% improvement over the extended Kalman filter (EKF)-based solution. The factor-graph approach exhibits significantly smoother trajectory behavior in challenging environments, confirming that the proposed robust framework offers superior accuracy and reliability with the high computational efficiency required for real-time applications.

Index Terms—PPP-B2b signal, inertial navigation system, factor graph optimization, integrated navigation.

I. INTRODUCTION

WITH the completion of the BeiDou-3 (BDS-3) constellation, the satellite-based augmentation signal PPP-B2b

Received 9 December 2025; revised 1 February 2026; accepted 9 February 2026. Date of publication 27 April 2026; date of current version 13 May 2026. This work was supported in part by the Young Scientists Fund of the National Natural Science Foundation of China under Grant 42404054 and in part by Ningbo Natural Science Foundation under Grant 2023J285. The associate editor coordinating the review of this article and approving it for publication was Prof. Yongchan Gao. (Corresponding author: Guangteng Fan.)

Ruite Yi is with the School of Systems Science and Engineering, Sun Yat-sen University, Guangzhou 510000, China, and also with the School of Electrical and Communication Engineering, Shenzhen Campus of Sun Yat-sen University, Shenzhen 518000, China (e-mail: yirt@mail2.sysu.edu.cn).

Xiangwei Zhu is with the School of Electrical and Communication Engineering, Shenzhen Campus of Sun Yat-sen University, Shenzhen 518000, China (e-mail: zhuxw666@mail.sysu.edu.cn).

Mingjun Ouyang is with the School of Electronics and Information Technology, Guangdong Polytechnic Normal University, Guangzhou 510000, China (e-mail: work_ouyang@163.com).

Chengchao Bai is with the School of Astronautics, Harbin Institute of Technology, Harbin 150001, China (e-mail: baichengchao@hit.edu.cn).

Guangteng Fan is with the Defense Innovation Institute, Academy of Military Sciences (AMS), Beijing 100000, China (e-mail: fanguangteng@163.com).

Digital Object Identifier 10.1109/LSP.2026.3687756

provides users with real-time high-precision orbit and clock corrections, which has greatly promoted global real-time precise point positioning (PPP) applications [1], [2], [3].

BDS-3 PPP-B2b broadcasts real-time corrections via geostationary earth orbit (GEO) satellites [4]. Its asynchronous update rates (6 s for clock, 48 s for orbit) and time-varying uncertainties necessitate a flexible fusion framework, making factor graph optimization (FGO) superior to traditional filtering. Recent studies have evaluated PPP-B2b product quality [5], [6], [7] and validated its real-time accuracy [8], [9], [10]. Comparisons with Galileo high accuracy service (HAS) [11] and multi-GNSS advanced orbit and clock augmentation (MADOCa)-PPP [12], alongside systematic benchmarking [13], [14], highlight its distinct service characteristics. However, challenges regarding long convergence [3] and insufficient urban robustness persist.

While tightly coupled PPP/inertial navigation system (INS) [15] mitigates signal outages, mainstream extended Kalman filter (EKF)-based approaches [15], [16] suffer from linearization errors and sensitivity to outliers. Furthermore, standard EKF fails to fully exploit cross-epoch constraints in complex real-time scenarios [17].

Alternatively, factor graph optimization (FGO) [18] employs sliding-window smoothing to enable iterative relinearization and robust weighting [19], [20]. In global navigation satellite system (GNSS)/INS contexts, FGO has been shown to improve urban reliability [21], [22], robustness against abnormal observations [23], and availability in multi-source fusion [24].

Nevertheless, dedicated modeling and implementation for PPP-B2b/INS remain limited. This letter proposes a factor-graph-based PPP-B2b/INS framework (Fig. 1) to address these limitations. The main contributions of this letter are: 1) a factor-graph-based tightly coupled PPP-B2b/INS framework that leverages iterative relinearization to mitigate linearization errors; and 2) a time-varying variance model with Huber loss to explicitly handle correction latency and signal degradation.

II. METHODOLOGY

This section summarizes the tightly coupled PPP-B2b/INS state and system models, the PPP-B2b-based GNSS observation model, and the factor-graph adopted in this work.

A. PPP Based on PPP-B2b Corrections Observation Model

For each satellite s , dual-frequency code and carrier-phase measurements on frequencies f_1 and f_2 are combined into ionosphere-free (IF) observables. Denoting code as $P_{IF,k}^s$ and

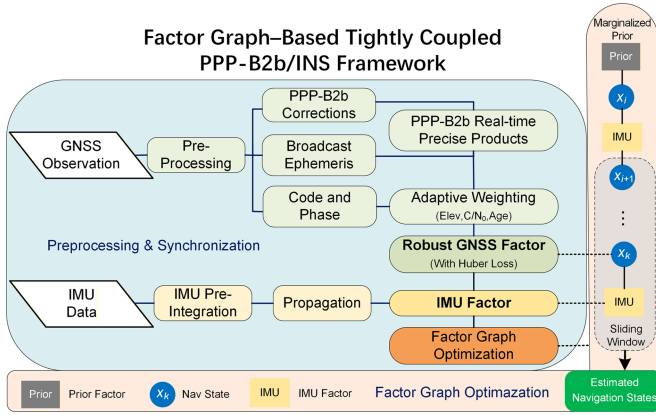


Fig. 1. Factor-graph-based tightly coupled PPP-B2b/INS integration framework.

phase as $\Phi_{IF,k}^s$ on frequency f_i , the IF combinations are:

$$P_{IF,k}^s = \alpha_1 P_{1,k}^s + \alpha_2 P_{2,k}^s, \quad (1)$$

$$\Phi_{IF,k}^s = \alpha_1 \Phi_{1,k}^s + \alpha_2 \Phi_{2,k}^s, \quad (2)$$

where $\alpha_i = f_i^2 / (f_1^2 - f_2^2)$ and the corresponding IF wavelength is λ_{IF} .

BDS-3 PPP-B2b orbit and clock corrections are applied to the broadcast ephemeris to obtain corrected satellite positions $r_k^{s,*}$ and clocks $\delta t_k^{s,*}$. Let $\rho(r_k, r_k^{s,*})$ denote the geometric distance between the receiver and satellite, T_k the slant tropospheric delay, and b_P^s, b_Φ^s the remaining code and phase hardware delays. The PPP-B2b-provided satellite DCBs (referenced to B3I signal) are applied to the raw code measurements before forming the ionosphere-free observables. The observation equations for the IF code and phase at epoch k are modeled as

$$P_{IF,k}^s = \rho(r_k, r_k^{s,*}) + c(\delta t_k - \delta t_k^{s,*}) + T_k + b_P^s + \varepsilon_{P,k}^s, \quad (3)$$

$$\begin{aligned} \Phi_{IF,k}^s &= \rho(r_k, r_k^{s,*}) + c(\delta t_k - \delta t_k^{s,*}) + T_k \\ &+ \lambda_{IF} N_k^s + b_\Phi^s + \varepsilon_{\Phi,k}^s, \end{aligned} \quad (4)$$

where N_k^s is the IF ambiguity, and $\varepsilon_{P,k}^s, \varepsilon_{\Phi,k}^s$ are zero-mean observation errors. The information from the removed observations and states regarding this ambiguity is compressed into the prior factor, which imposes constraints on the remaining states and ambiguities.

In order to address the asynchronous update intervals of the PPP-B2b corrections (48 s for orbit and 6 s for clock), we employed the Hold-Last method to align these corrections with the 1 Hz GNSS data. In this approach, the most recent available PPP-B2b corrections are retained. However, considering that satellite clock errors exhibit significantly faster temporal variations than orbit errors, the stochastic model is primarily driven by the latency of the clock corrections. Therefore, we define the correction age Δt_{age} specifically as the time elapsed since the last received clock correction:

$$\Delta t_{age} = t_{now} - t_{last_clock} \quad (5)$$

where t_{now} is the current GNSS epoch and t_{last_clock} is the time tag of the most recent PPP-B2b clock update. This ensures

that the variance inflation in the factor graph accurately reflects the rapid degradation of clock estimates over time, which is the dominant error source in real-time processing.

To ensure the reproducibility of the experiment, the stochastic model explicitly accounts for the noise amplification of the IF combination and the time-varying quality of PPP-B2b corrections. The variance σ_{IF}^2 for the IF measurement of a satellite at elevation θ is modeled as

$$\sigma_{IF}^2 = \mathcal{R}_{type}^2 \cdot \left(a_\sigma^2 + \frac{b_\sigma^2}{\sin^2 \theta} \right) + (\dot{e}_{clk} \cdot \Delta t_{age})^2, \quad (6)$$

where a_σ and b_σ are empirical base noise constants (set to 0.003 m for carrier phase); $\mathcal{R}_{type}$ is the noise ratio factor, set to 1 for carrier phase and 100–500 for pseudorange measurements to reflect the lower precision of code observations; and $\dot{e}_{clk}$ represents the uncertainty growth rate of the satellite clock (empirically set to 0.002 m/s). This function ensures that the weight of the observation automatically degrades as the satellite elevation decreases or the correction latency increases.

The resulting variances are assembled into a diagonal covariance matrix $\mathbf{R}_{GNSS,k}$ for each epoch, which is used in the GNSS factors of the graph and forms a key component of the proposed robust factor-graph-based PPP-B2b/INS integration.

B. State and INS Models

We use a standard error-state formulation for tightly coupled GNSS/INS integration. At epoch k , the navigation state is defined as

$$x_k = [r_k^T \ v_k^T \ \theta_k^T \ b_{g,k}^T \ b_{a,k}^T \ \delta t_k \ d_{T,k} \ N_k^T]^T, \quad (7)$$

where r_k and v_k are the receiver position and velocity in the local navigation frame, θ_k is a minimal attitude representation (e.g., roll, pitch, yaw), $b_{g,k}$ and $b_{a,k}$ are gyroscope and accelerometer biases, δt_k is the receiver clock offset, $d_{T,k}$ is the zenith tropospheric delay, and N_k stacks the carrier-phase ambiguities for all tracked satellites and arcs.

Let $\omega m, k$ and bg, k denote the raw inertial measurement unit (IMU) measurements of angular rate and specific force at time k . After bias compensation, the effective inputs for strapdown mechanization are

$$\tilde{\omega}k = \omega m, k - bg, k, \quad (8)$$

$$\tilde{f}k = fm, k - bm, k. \quad (9)$$

The continuous-time INS dynamics can be written compactly as

$$\dot{x}k = f(xk, \tilde{\omega}k, \tilde{f}k) + wk, \quad (10)$$

where $f(\cdot)$ collects the standard strapdown inertial equations together with random-walk models for sensor biases, clock offset and troposphere, and wk is zero-mean Gaussian process noise.

In the factor graph, a high-rate stream of IMU samples between two key epochs is summarized by an IMU pre-integration factor. For two consecutive keyframes $k-1$ and k , all raw IMU measurements in $(t_{k-1}, t_k]$ are integrated to obtain a relative motion constraint with mean and covariance that depend on

the continuous INS model above. This significantly reduces the number of variables and allows us to handle high-rate IMU (e.g., hundreds of Hz) together with lower-rate PPP-B2b observations in a unified optimization framework.

C. Factor Graph Optimization

Let $X = \{x_0, x_1, \dots, x_K\}$ denote the set of states within the current time window and Z the collection of all IMU and PPP-B2b measurements in this window. Under standard Gaussian assumptions, the posterior density factors as

$$p(X|Z) \propto f_0(x_0) \prod_k f_{IMU}(x_{k-1}, x_k) \prod_{k,s} f_{GNSS}^s(x_k), \quad (11)$$

where $f_0(\cdot)$ is the prior factor on the initial state, $f_{IMU}(x_{k-1}, x_k)$ represents the IMU pre-integration constraint between two keyframes, and $f_{GNSS}^s(x_k)$ collects the code and phase measurements from satellite s at epoch k .

Maximizing this posterior is equivalent to minimizing a non-linear least-squares cost function of the form

$$J(X) = \|r_0\|_{\Sigma_0^{-1}}^2 + \sum_k \|r_k^{IMU}\|_{\Sigma_{IMU,k}^{-1}}^2 + \sum_{k,s} \rho\left(\|r_k^{GNSS,s}\|_{\mathbf{R}_{GNSS,k}^{-1}}^2\right), \quad (12)$$

where r_0 is the prior residual, r_k^{IMU} is the IMU pre-integration residual between x_{k-1} and x_k , $r_k^{GNSS,s}$ stacks the code and phase residuals for satellite s at epoch k ; Σ_0 , $\Sigma_{IMU,k}$ and $\mathbf{R}_{GNSS,k}$ are the corresponding covariance matrices, and $\rho(\cdot)$ is a robust loss function applied to the GNSS residuals.

In this work, a Huber kernel is used as $\rho(r_{w,i})$ to suppress the influence of multipath and outliers while keeping small residuals almost unchanged. In the optimization, for the i -th whitened residual component $r_{w,i}$, the Huber cost function is defined as

$$\rho(r_{w,i}) = \begin{cases} \frac{1}{2} r_{w,i}^2 & \text{if } |r_{w,i}| \leq \delta \\ \delta(|r_{w,i}| - \frac{1}{2}\delta) & \text{if } |r_{w,i}| > \delta \end{cases}. \quad (13)$$

III. EXPERIMENTS

A. Experimental Setup

To assess the effectiveness of the proposed factor-graph-based PPP-B2b/INS integration, one representative dataset was selected from the open-source GICI-LIB project [22]. The dataset contains synchronized dual-frequency GNSS and MEMS-IMU observations collected under different environmental conditions. Dataset 5.1 was recorded in a dense urban environment with frequent signal blockage and strong multipath. A typical scenario is shown in Fig. 2.

GNSS measurements were processed at 1 Hz, IMU data at 400 Hz. The PPP-B2b correction was decoded by Sino K803S receiver and aligned, and robust kernel weighting together with time-varying measurement covariance was applied within each optimization iteration. For comparison, a conventional EKF-based tightly coupled PPP-B2b/INS solution was implemented using the same GNSS/IMU data and PPP-B2b corrections.



Fig. 2. Typical scenario of the dataset.

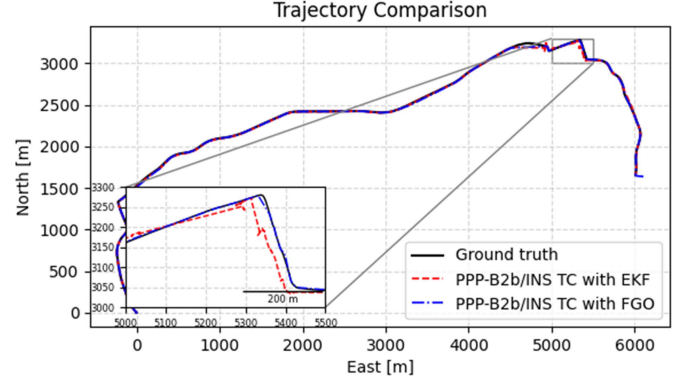


Fig. 3. Horizontal trajectory comparison between the EKF-based and FGO-based tightly coupled PPP-B2b/INS solutions.

In this experiment, a high-grade fiber optic IMU was rigidly mounted on the platform and used to generate the reference trajectory. The fiber IMU data and GNSS raw measurements were post-processed in a tightly coupled GNSS/INS scheme with forward-backward smoothing. The resulting solution provides centimeter-level position and high-accuracy attitude, and is treated as the ground truth for evaluating the real-time PPP-B2b/INS algorithms.

B. Algorithm and Parameter Settings

We employ a sliding-window factor graph optimization framework with a window size of 3 keyframes. Given the 1 Hz GNSS update rate, this corresponds to a time window of 3.0 seconds. States outside the window are marginalized at each update, so the number of states and factors per optimization remains bounded, resulting in bounded per-epoch computational cost. A Huber loss with threshold $\delta = 1.5$ is applied to the whitened GNSS residuals.

To ensure a fair comparison, the baseline EKF incorporates innovation-based outlier detection and IGG III robust filtering [25]. Thus, the study effectively compares “Robust EKF” against the proposed “Robust FGO.” The performance gains are attributed to FGO’s iterative relinearization and sliding-window optimization capabilities, which offer superior handling of non-linearities compared to the recursive EKF, even when both utilize robust kernels.

C. Results on Dataset 5.1

Fig. 3 compares the horizontal trajectories obtained from the proposed PPP-B2b/INS tightly coupled solutions with EKF

TABLE I
RMS POSITIONING ERRORS OF THE PROPOSED FACTOR-GRAPH-BASED PPP-B2B/INS SOLUTION

RMSE (m)	East	North	Up	3D
Model				
EKF	0.797	0.954	1.616	2.039
FGO (Proposed)	0.490	0.671	0.797	1.151

and FGO against the ground truth. At the global scale, both tightly coupled solutions follow the overall driving path and reproduce the main turns and straight segments, indicating that the PPP-B2b corrections and the INS mechanization are properly integrated. However, noticeable discrepancies appear when inspecting the local enlarged view.

The zoomed-in inset highlights a representative segment around a sharp corner with dense tree cover where satellite visibility drops significantly. In this challenging section, the EKF-based solution (the red dashed trajectory) exhibits a clear cross-track bias with respect to the ground truth. In contrast, the FGO-based solution (blue dash-dotted line) remains almost superposed with the reference trajectory throughout the straight segment and closely. These observations indicate that the factor-graph formulation can better exploit the temporal correlation of the INS states and the redundancy of multi-epoch PPP-B2b measurements, thereby constraining the trajectory more effectively under rapid dynamics and GNSS geometry changes. As a result, the FGO-based tightly coupled PPP-B2b/INS solution provides improved horizontal consistency compared with the conventional EKF implementation.

In the converged period, the EKF solution attains RMS errors of 0.797 m, 0.954 m, and 1.616 m in the east, north, and up components, respectively, whereas the factor-graph-based solution reduces the RMS values to 0.490 m, 0.671 m, and 0.797 m, as shown in Table I. This corresponds to improvements of about 38% (east), 30% (north), and 51% (up) with respect to the EKF. The improvement is most pronounced in the vertical component, where the factor graph effectively suppresses the large bias and long-term drift observed in the EKF trajectory.

Overall, these results indicate that the proposed factor-graph-based PPP-B2b/INS integration provides faster convergence, smaller steady-state errors, and more stable behavior in all ENU directions compared with the conventional EKF-based scheme.

D. Computing Efficiency and Real-time Performance

To validate the real-time feasibility of the proposed scheme, we evaluated the computational efficiency of both the EKF-based and FGO-based algorithms on a standard PC equipped with an Intel Core i9-14900HX CPU. The average execution times per epoch for the GNSS/INS integration (at 1 Hz GNSS update rate) are summarized in Table II.

The EKF-TC consumes approximately 4.2 ms per epoch, benefitting from its recursive single-step update structure. In comparison, the proposed FGO scheme requires an average

TABLE II
COMPUTATIONAL EFFICIENCY COMPARISON (PER EPOCH)

Algorithm	Solver	Max Iterations	Window Size	Avg. Time (ms)	Real-time Status
EKF-TC	Recursive Filter	1	N/A	4.2	YES
FGO-TC	L-M (Dense Schur)	5	3s	45.6	YES

of 45.6 ms per epoch. This increased computational load is expected and stems from the FGO framework's specific design: 1) Multiple relinearizations: unlike the single linearization in EKF, the FGO solver performs up to 5 Levenberg-Marquardt iterations per epoch to iteratively correct linearization errors; and 2) Batch Optimization: The solver jointly optimizes states within a 3.0-second sliding window rather than a single state vector.

Despite the higher computational cost compared to EKF, the FGO processing time (45.6 ms) is significantly lower than the GNSS data sampling interval (1000 ms), leaving a substantial margin (over 95% idle time) for other tasks. Furthermore, as the sliding window moves, old states are marginalized, keeping the size of the optimization graph constant. This ensures that the computational cost remains bounded and does not grow over time. Consequently, the proposed FGO framework achieves a favorable trade-off, verifying its capability for online real-time applications while delivering the better accuracy and robustness demonstrated in Section III-C.

IV. CONCLUSION

This Letter investigated a factor-graph-based tightly coupled PPP-B2b/INS integration scheme for real-time precise positioning with BDS-3 PPP-B2b corrections. GNSS measurements and IMU pre-integration were jointly modeled in a factor graph with robust loss functions and time-varying measurement covariance, so that multipath, signal degradation and correction quality could be handled while maintaining real-time computational efficiency. Using one representative dataset from the GICI-LIB, the proposed method consistently outperformed a conventional EKF-based tightly coupled PPP-B2b/INS solution, reducing the converged RMS position errors from 0.797/0.954/1.616 m to 0.490/0.671/0.797 m in the east, north and up components, respectively. These results demonstrate that the factor-graph formulation offers faster convergence, higher steady-state accuracy and better robustness under satellite masking and correction interruptions, making it a promising alternative to traditional filtering for real-time PPP-B2b/INS applications.

ACKNOWLEDGMENT

The authors would like to thank Dr. Cheng Chi from Shanghai Jiao Tong University for open-sourcing the GICI-Open-Dataset and codes.

REFERENCES

- [1] J. Tao, J. Liu, Z. Hu, Q. Zhao, G. Chen, and B. Ju, "Initial assessment of the BDS-3 PPP-B2b RTS compared with the CNES RTS," *GPS Solutions*, vol. 25, no. 4, Oct. 2021, Art. no. 131, doi: [10.1007/s10291-021-01168-1](https://doi.org/10.1007/s10291-021-01168-1).
- [2] Y. Xu, Y. Yang, and J. Li, "Performance evaluation of BDS-3 PPP-B2b precise point positioning service," *GPS Solutions*, vol. 25, no. 4, Oct. 2021, Art. no. 142, doi: [10.1007/s10291-021-01175-2](https://doi.org/10.1007/s10291-021-01175-2).
- [3] Y. Yang, Q. Ding, W. Gao, J. Li, Y. Xu, and B. Sun, "Principle and performance of BDSBAS and PPP-B2b of BDS-3," *Satell. Navigation*, vol. 3, no. 1, Dec. 2022, Art. no. 5, doi: [10.1186/s43020-022-00066-2](https://doi.org/10.1186/s43020-022-00066-2).
- [4] China Satellite Navigation Office, "BeiDou navigation satellite system signal in space interface control document: Precise point positioning service signal PPP-B2b (Version 1.0)," Beijing, China: CSNO, 2020.
- [5] H. Zhou et al., "Multi-frequency BDS-3 real-time positioning performance assessment using new PPP-B2b augmentation service," *IEEE Sensors J.*, vol. 23, no. 5, pp. 4994–5002, Mar. 2023, doi: [10.1109/JSEN.2023.3235901](https://doi.org/10.1109/JSEN.2023.3235901).
- [6] Z. Ren, H. Gong, J. Peng, C. Tang, X. Huang, and G. Sun, "Performance assessment of real-time precise positioning using BDS PPP-B2b service signal," *Adv. Space Res.*, vol. 68, no. 8, pp. 3242–3254, Oct. 2021, doi: [10.1016/j.asr.2021.06.018](https://doi.org/10.1016/j.asr.2021.06.018).
- [7] W. Dai et al., "BDS-3/GPS uncombined real-time PPP with PPP-B2b precise products: Modeling and its long-term performance evaluation," *Adv. Space Res.*, vol. 75, no. 10, pp. 7212–7225, May 2025, doi: [10.1016/j.asr.2025.01.025](https://doi.org/10.1016/j.asr.2025.01.025).
- [8] T. Geng, Z. Li, X. Xie, W. Liu, Y. Li, and Q. Zhao, "Real-time ocean precise point positioning with BDS-3 service signal PPP-B2b," *Measurement*, vol. 203, Nov. 2022, Art. no. 111911, doi: [10.1016/j.measurement.2022.111911](https://doi.org/10.1016/j.measurement.2022.111911).
- [9] H. Zhou et al., "Real-time single-frequency precise point positioning using BDS-3 PPP-B2b corrections," *Measurement*, vol. 205, Dec. 2022, Art. no. 112178, doi: [10.1016/j.measurement.2022.112178](https://doi.org/10.1016/j.measurement.2022.112178).
- [10] C. Ouyang, J. Shi, W. Peng, X. Dong, J. Guo, and Y. Yao, "Exploring characteristics of BDS-3 PPP-B2b augmentation messages by a three-step analysis procedure," *GPS Solutions*, vol. 27, no. 3, Jul. 2023, Art. no. 119, doi: [10.1007/s10291-023-01457-x](https://doi.org/10.1007/s10291-023-01457-x).
- [11] N. Naciri, D. Yi, S. Bisnath, F. J. de Blas, and R. Capua, "Assessment of Galileo high accuracy service (HAS) test signals and preliminary positioning performance," *GPS Solutions*, vol. 27, no. 2, Feb. 2023, Art. no. 73, doi: [10.1007/s10291-023-01410-y](https://doi.org/10.1007/s10291-023-01410-y).
- [12] A. S. Pillai et al., "Performance evaluation of MADOCA PPP with existing positioning techniques for UAV altitude determination," in *Proc. Int. Conf. Comput. Intell. Appl.*, Jul. 2025, pp. 1–5, doi: [10.1109/CIA-CON65473.2025.11189728](https://doi.org/10.1109/CIA-CON65473.2025.11189728).
- [13] T. Willems et al., "Disseminating regional ionospheric corrections from GNSS constellations and application to Galileo high accuracy service," *GPS Solutions*, vol. 29, no. 4, Sep. 2025, Art. no. 194, doi: [10.1007/s10291-025-01955-0](https://doi.org/10.1007/s10291-025-01955-0).
- [14] D. Kiliszek, "GPS and Galileo precise point positioning performance with tropospheric estimation using different products: BRDM, RTS, HAS, and MGEX," *Remote Sens.*, vol. 17, no. 12, Jun. 2025, Art. no. 2080, doi: [10.3390/rs17122080](https://doi.org/10.3390/rs17122080).
- [15] W. Li, C. Pan, J. Gao, Z. Li, and Z. Tao, "Comprehensive evaluation of real-time uncombined-PPP/INS tightly coupled integration with BDS-3 PPP-B2b service in urban environments," *Meas. Sci. Technol.*, vol. 36, no. 2, Feb. 2025, Art. no. 026310, doi: [10.1088/1361-6501/ad8896](https://doi.org/10.1088/1361-6501/ad8896).
- [16] P. Xiao, F. Sun, K. Wang, K. Xiao, X. Shang, and J. Liu, "Positioning performance analysis of real-time BDS-3 PPP-B2b/INS tightly coupled integration in urban environments," *Adv. Space Res.*, vol. 72, no. 9, pp. 4008–4020, Nov. 2023, doi: [10.1016/j.asr.2023.08.056](https://doi.org/10.1016/j.asr.2023.08.056).
- [17] S. Xin, X. Wang, J. Zhang, K. Zhou, and Y. Chen, "A comparative study of factor graph optimization-based and extended Kalman filter-based PPP-B2b/INS integrated navigation," *Remote Sens.*, vol. 15, no. 21, Nov. 2023, Art. no. 5144, doi: [10.3390/rs15215144](https://doi.org/10.3390/rs15215144).
- [18] F. R. Kschischang, B. J. Frey, and H.-A. Loeliger, "Factor graphs and the sum-product algorithm," *IEEE Trans. Inf. Theory*, vol. 47, no. 2, pp. 498–519, Feb. 2001, doi: [10.1109/18.910572](https://doi.org/10.1109/18.910572).
- [19] Z. Yang, X. Ding, Y. Yang, and Q. Wang, "OiSAM-FGO: An efficient factor graph optimization algorithm for GNSS/INS integrated navigation system," *Satell. Navigation*, vol. 6, no. 1, Dec. 2025, Art. no. 23, doi: [10.1186/s43020-025-00173-w](https://doi.org/10.1186/s43020-025-00173-w).
- [20] H. Zhang, X. Xia, M. Nitsch, and D. Abel, "Continuous-time factor graph optimization for trajectory smoothness of GNSS/INS navigation in temporarily GNSS-denied environments," *IEEE Robot. Automat. Lett.*, vol. 7, no. 4, pp. 9115–9122, Oct. 2022, doi: [10.1109/LRA.2022.3189824](https://doi.org/10.1109/LRA.2022.3189824).
- [21] T. Li, H. Zhang, B. Han, M. Xia, and C. Shi, "Relative accuracy of GNSS/INS integration based on factor graph optimization," *IEEE Sensors J.*, vol. 24, no. 20, pp. 33182–33194, Oct. 2024, doi: [10.1109/JSEN.2024.3451665](https://doi.org/10.1109/JSEN.2024.3451665).
- [22] C. Chi, X. Zhang, J. Liu, Y. Sun, Z. Zhang, and X. Zhan, "GICI-LIB: A GNSS/INS/camera integrated navigation library," *IEEE Robot. Automat. Lett.*, vol. 8, no. 12, pp. 7970–7977, Dec. 2023, doi: [10.1109/LRA.2023.3324825](https://doi.org/10.1109/LRA.2023.3324825).
- [23] X. Xia, L.-T. Hsu, and W. Wen, "Integrity-constrained factor graph optimization for GNSS positioning," in *Proc. IEEE/ION Position Location Navigation Symp.*, Monterey, CA, USA, 2023, pp. 414–420.
- [24] S. Zhang, R. Tu, Z. Gao, D. Zou, S. Wang, and X. Lu, "LEO-enhanced GNSS/INS tightly coupled integration based on factor graph optimization in the urban environment," *Remote Sens.*, vol. 16, no. 10, May 2024, Art. no. 1782, doi: [10.3390/rs16101782](https://doi.org/10.3390/rs16101782).
- [25] Y. Yuanxi, "Robust estimation for dependent observations," *Manuscripta Geodetica*, vol. 19, no. 1, pp. 10–17, Feb. 1994, doi: [10.1007/BF03655325](https://doi.org/10.1007/BF03655325).